Homological Methods in Commutative Algebra

Problem Set 2

Throughout, all rings all commutative and noetherian, and all modules are finitely generated.

Problem 1. Let I be a nonzero proper ideal in a noetherian domain R and let $f_1, \ldots, f_c$ be a maximal regular sequence inside I. Consider the short exact sequence

$$0 \longrightarrow N \longrightarrow R/(f_1, \ldots, f_c) \xrightarrow{\pi} R/I \longrightarrow 0.$$

where π is the canonical quotient map.

a) Show that $\operatorname{Ext}_R^{c-1}(N, R) = 0$.

b) Show that the following induced map on Ext is nonzero:

$$\pi^* = \operatorname{Ext}_R^c(\pi, R) \colon \operatorname{Ext}_R^c(R/I, R) \longrightarrow \operatorname{Ext}_R^c(R/(f_1, \ldots, f_c), R).$$

Problem 2. Let $(R, \mathfrak{m})$ be a noetherian local ring and take elements $f_1, \ldots, f_c \in I$. Consider the koszul complex $E = \operatorname{kos}(\underline{f}) = \bigwedge(e_1, \ldots, e_c)$ and let F be a dg R-algebra resolution of R/I with $F_0 = R$. Fix $a_1, \ldots, a_c \in F_1$ such that $\partial_F(a_i) = f_i$. We define a map of complexes $\varphi \colon E \longrightarrow F$ as follows: we set $\varphi_0 = \operatorname{id}_R$ and for each $1 \leqslant s \leqslant c$, consider the R-module map $\varphi_s \colon E_i \longrightarrow F_s$ given on basis elements by

$$\varphi_s(e_{i_1} \wedge \cdots \wedge e_{i_s}) = a_{i_1} \cdots a_{i_s}.$$

a) Show that $\varphi\partial = \partial\varphi$, that is, that φ is a map of complexes.

b) Show that for all $z \in \ker(\varphi)$ and all $w \in E$, $wz \in \ker(\varphi)$.

Let k be a field and let $f_1, \ldots, f_n$ be monomials in $k[x_1, \ldots, x_d]$, minimally generating the ideal $I = (f_1, \ldots, f_n)$. The Taylor resolution of R/I has a dg algebra structure, defined on basis elements as follows:

$$e_J \cdot e_L = \operatorname{sgn}(J, L)\, \frac{f_J f_L}{f_{J \cup L}}\, b_{J \cup L}\,.$$

Here the sign is given by

$$\operatorname{sgn}(J, L) = (-1)^{\varepsilon} \quad \text{where} \quad \varepsilon = |\{(j, \ell) \,:\, j \in J,\ \ell \in L,\ \text{and } j > \ell\}|$$

and by convention, if $J \cap L \neq \varnothing$, then $\operatorname{sgn}(J, L) = 0$.

Problem 3. Let $I = (f_1, \ldots, f_n)$ be a monomial ideal in $Q = k[x_1, \ldots, x_d]$ and consider its Taylor resolution T. Fix a subset $J \subseteq [n]$.

a) Show that

$$\partial(e_J) \notin \mathfrak{m}T \iff f_i \mid f_{J \setminus \{i\}} \text{ for some } i \in J.$$

b) Show that for each $i \in [n]$,

$$e_i e_J \notin \mathfrak{m}T \iff \gcd(f_i, f_j) = 1 \text{ for all } j \in J.$$

Problem 4. Let k be any field and $R = k[x, y, z_1, \dots, z_n]$ for some $n \geqslant 3$. Show that the ideal

$$I = \left(x^n, y^n, \sum_{i=0}^{n-1} z_{i+1} x^i y^{n-i} \right)$$

has $\operatorname{pdim}(R/I) = n + 2$.